

SUSMA R

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SUMMARY

AI Engineer with hands-on experience in developing and deploying LLM-driven applications using Python, FastAPI, LangChain, Vector Search, and Cloud-native architectures. Skilled in translating business requirements into scalable AI solutions, collaborating with cross-functional teams, and implementing production-grade automation pipelines.

WORK EXPERIENCE

AI Engineer, RedBlackTree Technologies Pvt Ltd (Chennai)

Jan 2025 - Nov 2025

Project 1: Invoice Processing System

- Built and deployed a GenAI-powered invoice automation system using **Gemini**, **Langchain**, **LangGraph**, and **LangSmith** for workflow observability, performance tracking, and debugging.
- Automated key finance operations including supplier mapping, account/store assignment, and journal entry determination reducing manual intervention by **60%**.
- Integrated with **PostgreSQL** for reliable workflow state persistence, improving process traceability and system stability for enterprise adoption.
- Developed secure **FastAPI** microservices, improving invoice processing speed by **35%** and reducing financial errors by **30%** through automated validation checks.

Project 2: Facial Recognition Attendance Kiosk

- Developed an Android-based kiosk application using Kotlin + Jetpack Compose for **real-time facial recognition-based attendance**.
- Integrated facial embeddings and optimized similarity search using **pgvector** in **PostgreSQL**.
- Designed a **hybrid architecture** with Room DB for local data storage, WorkManager for background sync, and **FastAPI** backend for secure processing.
- Leveraged **AWS Kinesis** for streaming facial recognition events to ensure reliable real-time data ingestion and analytics in online mode and implemented robust workflows for employee enrollment, **on-device matching**, and fault-tolerant **offline punch recording** with sync fallback.

Data Scientist Internship, Bosch Global Software Technologies (Bangalore)

Jan 2024 - May 2024

- Designed and implemented an **End-to-End Robustness Evaluation Pipeline** to assess AI/ML model performance under Adversarial and real-world conditions.
- Utilized **GANs** for synthetic data generation, Adversarial Testing, and Randomized Smoothing to evaluate model stability and resilience.
- Integrated Out-of-Distribution Detection and Counterfactual Explanation techniques to assess robustness and interpretability.
- Built and deployed scalable **FastAPI** services for real-time evaluation and visualization of robustness metrics.

Project Trainee, Bosch Global Software Technologies (Coimbatore)

Jul 2022 - Nov 2022

- Automated data collection from the **JIRA server** for real-time ticket status tracking and key metrics, ensuring streamlined reporting.
- Developed an interactive **RNM Metrics Dashboard** using Plotly and Dash to visualize ticket priorities, productivity, and quality, enhancing decision-making.

PROJECTS

Smart AI Interview Evaluation and Coaching Platform

Developed an AI-powered interview analysis system to process transcripts, detect bias, and generate structured STAR feedback. Implemented a multi-agent CrewAI architecture with specialized agents for skills mapping, coaching, and summary generation. Built pipelines for file processing, data validation, report generation, and secure API integration. Created a scalable, production-ready solution converting raw interview data into actionable insights for HR teams.

Tech Stack : Python, Streamlit, Mistral AI, CrewAI

Legal Information Retrieval System

Developed and AI-powered system using Python and OpenAI to translate user queries into relevant legal information, integrating Llama Index, Chainlit, and a local vector database. Implemented tools for extracting key legal concepts and identifying corresponding clauses in updated legal texts, ensuring precision and relevance line.

Tech Stack : Python, OpenAI, Llama Index, Chainlit, Local Vector Database

SKILLS

Generative AI & LLM	LLMs, LangChain, LangGraph, Prompt Engineering, RAG (Retrieval-Augmented Generation), Multi-Agent Systems (CrewAI), Fine-Tuning, GANs
Programming & Frameworks	Python, SQL, FastAPI, Flask, Scikit-learn, PyTorch, TensorFlow, Streamlit, OpenCV, HuggingFace, NLTK
Cloud & Deployment	AWS (Kinesis, S3, EC2, Lambda), Azure, Model Deployment, API Gateway
Databases	PostgreSQL, VectorDB, MySQL, MongoDB, NoSQL, ChromaDB
Tools & Analytics	NumPy, Pandas, PowerBI, Tableau, Git, Docker

EDUCATION

M.Sc. Decision and Computing Sciences(Integrated), Coimbatore Institute of Technology 2024 (CGPA: 8.69)

PUBLICATIONS

- “A Real Time Web-Based Flask Application For Detection Of Facial Emotions Using Deep Learning Model” (Apr. 2023). In: *JASC*.
- “Mushroom Classification To Detect Their Edibility using ML Algorithms and Rapid Miner” (Apr. 2023). In: *IRJMETs*.
- “Comprehensive Framework for Robustness Evaluation on Numeric Data Classification” (June 2024). In: *ICCCNT 2024 held at IIT-Mandi, IEEE Electronics Packaging Society*.